

An implementation and analysis of deep learning models for the detection of wheat rust disease

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Abstract—Wheat is one of the essential food crops all over the world. In India, it is the largest growing winter cereal crop that contributes almost 14% to the world's food production. This crop is usually affected by various types of rust diseases. Rust is one kind of fungal disease, which occurred on wheat crop. The rust is characterized as leaf-rust (brown rust), stem-rust (black rust), and yellow rust (strip rust). Due to these types of diseases, a huge amount of loss may occur in the production of wheat crop. Therefore, it is required to early detect this disease during the growing stage. In the present study, we consider two classes of diseased plants and one for a healthy plant. Due to less amount of image dataset, we performed the image augmentation and generated the synthetic data from the original data. Various experiments are performed on different dataset sample sizes *i.e.*, 100, 500, 1000, 1500, and 2000. The weights for building the model have been considered as the same in ResNet50 and VGG16 transfer learning models. Moreover, to improve the accuracy of the model, we tune the hyper-parameters. The hyper-parameters used to train this model are: *batch_size* = 32, *optimizer* = "adam", *initial learning rate* = 0.01 with decay of 0.0001. After performing the experiments, the model produced appreciable results with the VGG16 model. The proposed approach has obtained the highest classification accuracy rate of 99.07% using the VGG16 model.

Index Terms—Deep Learning, Convolutional Neural Network, Wheat disease detection, Wheat crop disease classification

I. INTRODUCTION

Wheat is a winter cereal crop in India and it fulfills everyone's need for food at the Global level. The growth of the crop depends upon the region, climate change, soil quality, and also by monitoring the growth of plants at various time-span. It serves as the staple food for billion people of people all over the world. It contributes roughly 14% to the world's wheat basket and holds an 11% global share of the region under wheat cultivation [1]. India stands next to China in the world, first in the region and second in revenue. India's wheat requirement was largely met during the 1950s by imports from other countries. But as the Green Revolution came in 1968, there was a vigorous growth in wheat crop and in 2001 we produced 74 million tons of wheat, compared to 6 million tons during the 1950s. Now, the production of wheat is 93 million tons in India on an area of about 30 million/ha. The production can be increased by curing crops of various diseases.

A. Agricultural disease

Agricultural diseases affect crop productivity and are more likely to affect food availability. We can define plant disease as the change of a normal state to a diseased state that may modify or interrupts the vital functions of the plant. All plant species, both wild and cultivated, suffer from the disease. Pathogenic agents such as fungi, viruses, and bacteria are responsible for the plant disease and Pests are living agents such as insects, rodents, cockroaches, fleas, flies, and so on. For example, Rodents can damage the food and transmit the disease to humans as well. These diseases may be suffered from biotic and abiotic stresses as given below:

a) Biotic factors: These are living things such as animals, plants, fungi, bacteria, etc and they also affect the ecosystem. They can spread from plant to plant and may infect all types of leaves, stems, crowns, roots, fruits, and seeds. These are called pathogens of plants. Biotic services comprise all living elements of an ecosystem.

b) Abiotic factors: Non-biotic factors include non-living things like forest fires, water, atmosphere, weather, soil, minerals, and human activity. This type of disease can not spread from plant to plant but its effect on plant health can be seen directly. The abiotic factors apply to all the non-living components of an environment. These are the physical conditions such as various temperatures required for various crops, pH value and humidity level, and chemical agents such as gases, nutrients, and minerals in the water, air, and soil.

B. Wheat rust:

Among all the diseases, rust is a common fungal disease which is occurred on wheat plants. Mainly, three types of rust are found on wheat plants. (a) stem or black rust which is caused by *Puccinia graminis* f. sp. *tritici* West. This type of disease has occurred in the southern region. (b) leaf or brown rust is caused by *Puccinia triticina* Eriks, and is commonly found in the northern area. (c) stripe or yellow rust caused by *Puccinia striiformis* f. *tritici* Eriks, which is also found in northern areas. The description is as shown in Table I.

(i) Stem rust: It is one of the types of rust disease which is occurred by a biotic factor *i.e.*, fungal infection. This type of disease is caused in central, south areas in India. Firstly, this disease occurred in 1786 A.D. in central India [2]. Various

reports of losses are reported due to the epidemics of stem rust. Globally, it is a significant threat to global production of wheat [3]. It occurred in the summer season therefore it is also known as summer rust. Initially, black teliospores were developed, and at the end of the season, uredinium formed on wheat plants. The favorable condition for the growth of this disease is a humid temperature of 15 to 35 degree Celsius and if it found its favorable temperature then it generated 50% loss in only one single month [4].

(ii) Leaf rust: Leaf rust is the most competitive of all the rusts. The occurrence of large-scale leaf rust was observed in the northern plains during 1971–1973 and in the northeastern plain region (NEPZ) during 1993–1994 [5]. At the end of the season, it also lacks the abundant teliospore development of stem rust and changed the leaf color to brown spores rather than black which are the identification of stem rust. Leaf rust teliospores were formed first in the lower surface and the favorable temperature for growth of this disease is between 10 to 30 degree Celsius [4].

(iii) Yellow (or stripe) rust: Stripe rust is found in the northern side of India and it is destructive and important. It can cause yield losses of up to 100 percent. Whenever it starts growing the yellow color urediniospores were found on wheat leaf and if it found its favorable temperature then it spread in other plants also [6]. The favorable temperature for its growth is between 2 to 15 degree Celsius [4]. Strip rust is a cooler climate disease, so new high-temperature-resistant species have recently been reported that considerably enhances the potential for damage [7]. It takes its name from the characteristics strip of uredinia which produces urediniospore in yellow color. Due to this disease, the early attack also occurs on the weakened plant.



Fig. 1. Sample of healthy wheat plants



Fig. 2. A sample wheat leaf rust

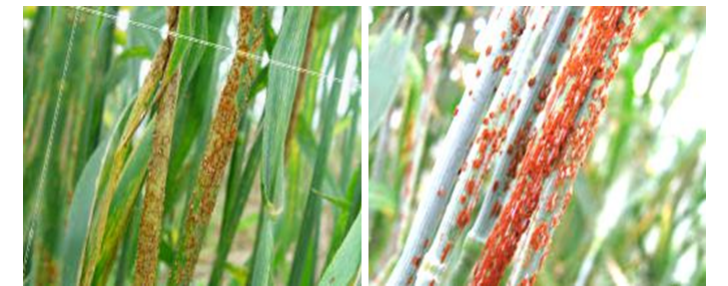


Fig. 3. A sample of wheat stem rust

TABLE I
SUMMARIZATION OF DISEASE AND IT SYMPTOMS [4]

Disease	Pathogen	Symptoms
Leaf rust	<i>Puccinia recondita</i> f.sp. tritici	Uredinia occurs on upper leaf surface and also sometimes on leaf sheath.
Stem rust	<i>Puccinia graminis</i> f.sp. tritici	Uredinia occurs on upper and lower leaf surface stem and spikes.
Strip rust	<i>Puccinia striiformis</i> f.sp. tritici	Systemic uredinia occurs first on leaves and then on spikes, sometimes leaf sheaths.

Fig. 1 contains the sample images of healthy wheat leaves. Fig. 2, Fig. 3 and Fig. 4 are the sample images of leaf rust, stem rust and strip rust on the wheat plants. In this article, the author has used the secondary dataset which has three classes such as healthy wheat, leaf rust, and stem rust as shown in Fig. 1, Fig. 2, and Fig. 3 respectively.

C. Life cycle of wheat rust disease

Wheat rust is called an obligate parasite means that it needs a host plant to complete its life cycle and is also called heteroecious fungi because it needs two host plants for completion of the life cycle. A primary host plant is wheat and



Fig. 4. A sample of wheat strip rust

a secondary host plant is barberry. The life cycle of *Puccinia* is divided into five stages, the first three stages occur on wheat plant and the last two stages occur on barberry plant.

Stage 1: Uredial stage, In the first stage aecidiospores germinated on wheat, and later dikaryotic hyphae will be formed on uredia bearing uredispores. Uredispore are unicellular structure shape and has unicellular structure and two nuclei are present inside it and they called diakaroyotic structure and complete

on the wheat plant. This stage is also called the red rust stage.

Stage 2: Telial stage, Seasonal change triggers the formation of telia or teleutospore. The main difference between uredial and telia stage is that uredispore is an unstructured stage and teleutospore are bicelled structure. Two cells are present and then they fuse and made diploid nuclei. Using meiosis these diploid nuclei formed four haploid nuclei. Two are positive and two will be negative.

Stage 3: Basidial stage, In this stage Basidiospore is formed from 4 haploid nuclei and further four basidiospores are formed, 2 are positive and 2 are negative.

Stage 4: Pycnidial stage, The basidiospore dispersed through the air on barberry plant. The mycelium is formed on a Barberry leaf that is called monokaryotic. This mycelium goes through upper epidemic into flower shape structure that is called pycnidium.

Stage 5: Aecial stage, This phase is also called diakaroyotic phase. In this phase, aecidiospore is formed and this aecidiospore is unicellular. It is dispersed by air and germinates on the primary host plant *i.e.*, the wheat plant to complete its life cycle.

D. Deep learning:

Nowadays, the use of deep learning for disease classification and detection is on the hike. Automatic wheat disease detection is useful and it validates to monitor the large crop fields. Automatic disease detection can be possible through the system which can detect disease at an initial level as they appear on wheat leaves. So that, necessary action can be taken in an early stage. Therefore, for improving the quality and quantity of crops there is a need to protect and detect in early-stage [8]. Therefore, by selecting the right medication and preventing the disease from spreading, having primary identification and pointing out the appropriate disease is more effective [9]–[11]. Deep learning is part of machine learning, which provides valuable results using computer vision techniques. To perform the experiments using deep learning models requires a huge dataset. A large amount of data produce efficient results *i.e.*, means our model will be ready for unseen data and produced the best results. Large quantities of data are produced on daily basis. Using this data we can perform experiments and train the deep learning model. There are two basic requirements for building deep learning models. One is a large amount of data and the other is efficient computation power such as Graphics Processing Unit(GPU) and High-Performance Computational (HPC) that make it possible to train deep models and to use computing parallelism. Nowadays, Convolutional Neural Network (CNN) is best for image classification and plant disease detection. Thus, features are extracted whenever a convolution operation is performed between the input image matrix and filter matrix. Applying filter on the image will produce features values from lower level to higher level *i.e.*, horizontal edge detection, vertical edge detection, and at a higher level, prominent features will be extracted from the images that will help to classify the image with a true class label.

II. LITERATURE SURVEY

Diagnosis of the disease earlier detected by human beings that prone to error. The author proposed a novel technique for automatic disease detection [12], in their study, they implemented a technique based on computer vision. Modern computer vision techniques solved the disease detection problem that will early detect the disease in the early stage of growth. The author created their dataset of images from two sites. One was Spain and the other was Germany in the various time-span 2014, 2015, and 2016 from various mobile devices. They implemented a novel algorithm called hot spot detection with statistical inference and the performance of the model was checked on the AUC curve metrics, and the resultant value achieved from greater than 0.80 for detecting the disease. An application for counting the urediospores for wheat strip rust developed by [13]. The experiment was conducted on a microscopic digital image on MATLAB. The authors solved the above-said problem in the following parts: (a) The input image has taken from counting spores. (b) Various image pre-processing techniques were used such as clustering segmentation, morphological modification, and also watershed segmentation used in this study. (c) As a result, the system was capable of counting the urediospores with the highest accuracy of 95%. A novel technique for crop disease has been proposed in [14] that can directly affect the reduction of production and also responsible for crop loss. The author has solved the given problem by identifying and localization of disease directly on the image dataset. The dataset used for the study is collected in filed in 2017 named as Wheat disease database (WDD2017). In this author has performed experiments using VGG-FCN-S, VGG-FCN-VD16 accuracy achieved 95.12% and 97.95% respectively. After achieving accuracy author deployed the proposed model as a mobile app. This was very helpful in the diagnosis of the disease. The author in [15] solved the problem of crop disease detection on wheat crop. In their study they used to consider four categories of wheat crops: among these four classes, three classes are diseased classes and one healthy class. They implemented the CNN based models with the combination of transfer learning model *i.e.*, AlexNet and achieved the accuracy of 84.54% for predicting the correct class. The author [16] proposed the novel technique for detecting the fungal disease on wheat crop in European endemic categories such as Rust, Tan spot, and Septoria. The dataset collected for this study has total images of 8,178 collected from various mobile devices in the year 2014, 2015, and 2016. They extend the work of [12] and used a novel technique of Deep Residual Neural Network algorithm of Deep learning. They achieved the accuracy greater than 96% for the experiment which were performed in Germany. The study classifies the ripened and non-ripened Dragon fruit using a deep learning model *i.e.*, ResNet152 [17]. The results of the experiments were compared with the VGG16/19 model with the measure of accuracy and loss. The experiments were performed with a variant of 10 to 500 epochs. The author analyzes the need

for Artificial Intelligence while buying fruits and vegetables. Everything in the supermarket is enclosed with the bar code for generating the invoice whereas fruits and vegetables are not enclosed with the bar code. The proposed system is validated with the accuracy on the classifiers Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbor (KNN), and Discriminant Analysis (DA) [18]. Among this classifier, KNN gives a prominent accuracy of 93.10%. The existing deep learning models in literature gives prominent result on image data. The advantage of using deep learning models is that it will automatically extract features from the input images while passing through the various layers whereas while increasing the number of layers the complexity of the model will be increased and it requires more time for training the model.

III. DATASET

The dataset is taken online from the Kaggle website and data for the images comes from several sources. The bulk of the data was collected in-field in Ethiopia and Tanzania by CIMMYT and CIMMYT partners. The majority of the data comes from public photos that are found on Google Images. The goal of this dataset is to create a model of machine learning to properly classify the wheat in the pictures as a healthy leaf, stem rust, or leaf rust. After download, first, we apply CNNs based model and found that the model has given very bad results on training as well as test data. The dataset we have considered for this study is very noisy data because the images are collected from different sources such as Google images and some of the few images are collected directly from the field and also to implement deep learning-based models we require a huge amount of data. So, to generate more data from existing data the technique called Image augmentation used, with the help of image augmentation we generated the images by varying various parameters as shown below:

A. Image augmentation

Image augmentation is the technology that helps to increase the dataset for performing the deep learning experiments. In another word, we can say that image augmentation is the change in training data that can be generalized to a certain extent. For example, image augmentation can be performed by cropping the image, rotating at a particular angle, shearing the image, reducing dimensions, and also shift according to width and height. Adjusting the brightness, changing color intensity *i.e.*, RGB to grayscale will produce the new dataset of images. Image augmentation is the part of pre-processing the images that are normalizing the image at the same dimension. It is only required in two conditions. Firstly, when less amount of dataset is available and also if the dataset contains noisy images. Secondly, to scale down all images to a fixed size for achieving better results. An important question to be answered is why increasing the dataset size increases accuracy? The answer to this question is that deep learning models give better results if the size of the dataset is large. From the big data, various features are extracted from various layers so based on those features our model

will better classify the input data into the correct class. In this study, we will see the effect of increasing the dataset to the accuracy of the model. The overfitting can be reduced by increasing the dataset for convolutional neural networks. The dataset considered for this study is not large enough so we perform the data augmentation as given below:

ImageDataGenerator: *ImageDataGenerator()* function is used in image augmentation for generation of batches. It has various parameters such as rotation of the input image at the particular angle which will change the intensity value of the image, shifting of image either width wise or height wise, shearing is the shifting one part of the image, *etc.* The description of the parameter is as given:

featurewise_center=True, featurewise_std_normalization=True, rotation_range=30, width_shift_range=0.2, height_shift_range=0.2, shear_range=0.2, zoom_range=0.1, fill_mode='reflect', brightness_range = (0.5, 1.5).

A description of these parameters is given below.

- (i) "*featurewise_center=True*", this parameter sets the means value to zero for all the inputs over the dataset and features.
- (ii) "*featurewise_std_normalization=True*", This parameter normalize all the data inputs and features in a standard range.
- (iii) "*rotation_range=30*", is used to set the random degree of rotation of dataset and also the value given to this attribute is integer value.
- (iv) "*width_shift_range*", it is used to set the width shift range of image means for what value it will shift the input image
- (v) "*height_shift_range*", is used to shift the input image according to the height of image.
- (vi) *shear_range*, this parameter stores float value and shear intensity (Shear angle in counter-clockwise direction in degrees).
- (vii) "*zoom_range*", Float or [lower, upper] is the range for random zoom.
- (viii) "*fill_mode*" has various parameters such as "constant", "nearest", "reflect" or "wrap". Default is 'nearest'. Points outside the boundaries of the input are filled.
- (ix) "*brightness_range*", it takes the input either by tuple or list in floats value. One is the start value and another is maximum value of the brightness. Using these parameters we have generated 2000 images per class from original dataset of wheat rust II.

TABLE II
GENERATION OF SYNTHETIC IMAGES FROM ORIGINAL IMAGES

Classes	Real images	Augmented data
Healthy wheat	142	2000
Leaf rust	358	2000
Stem rust	376	2000

1) *Proposed model:* In the present study, three channels (Red, Green, and Blue) colored images have been considered. First, the images are processed into the Convolution Neural Networks (CNN) based models such as ResNet50 and VGG16. These are the pre-trained models, which provide

the various layers and also weights associated with them. These weights are act as input for building the model and also these models have already implemented the convolution and max-pool layers with *ReLU* activation function. *ReLU* removes the non-linearity from the data. The combination *conv-pool* many times gives various features from the input image. After extracting features there are dense layers/fully-connected layers, the ResNet50 and VGG16 models classify the images of 1000 classes whereas in this study the author used 3 classes as an output class. Therefore, in the last layer using *softmax* function is used to change the output classes. The current study of the models ResNet50 and VGG16 is shown in Fig. 5.

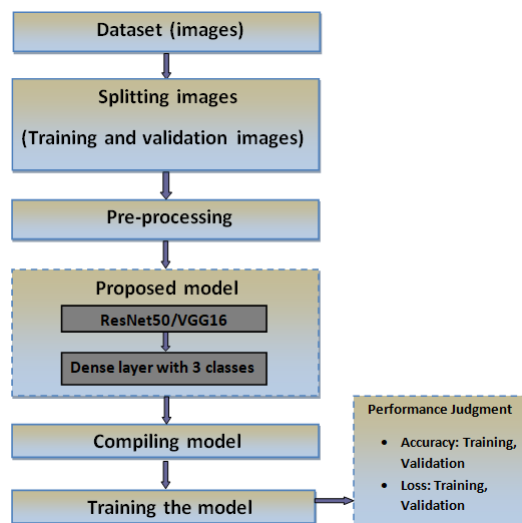


Fig. 5. Proposed model using (ResNet50/VGG16)

B. Methodology

In the modern era, Convolution Neural Networks (CNNs) are more widely used for image classification with promising accuracy. It is the family of neural networks that are designed for image classification, segmentation of the image, object detection, and also for neural style transfer. Building CNN's involves the following operation such as convolution, max-pool, ReLU (activation function) for feature extraction. After extracting the features, these are fed into the flatten layer, and the flatten layer passes these features to fully connected layer(s). The presented model in the present study is trained on the extracted features and also using the softmax function at the last layer. In this model, we have three class-labels such as leaf rust, stem_rust, healthy_wheat. To further improve the model accuracy, tuning with hyper-parameters is necessary. The hyper-parameters used to train the model are learning rate, optimizer, loss function plays a very important role in deep learning-based models. In this study, two neural networks architectures have been used to train the model with wheat rust dataset as discussed below:

(a) ResNet50: It is also one of the popular techniques of transfer learning. In this architecture, the input layer dimension

is used as $224 \times 224 \times 3$. Initially, convolution and max-pooling 7×7 and 3×3 kernel sizes, respectively. The use of these types of models is to initialize weights to a new model for further training using a different dataset. As the depth of the model is increased by integrating the new dataset, therefore, the new model may suffer from vanishing and exploding the gradient problem. *i.e.*, during training our model will not converge to a particular value. Sometimes, there is a huge difference between training and testing error that shows that our model is not optimized enough for the given dataset. ResNet50 model used the skip connections that are used to copy the layers from the existing layer. Skip connection provides the way for calculating the gradient for model optimization. Backpropagation is used to reduce the error while training the model, the backpropagation algorithm will help to adjust the weights in the model.

(b) VGG16: It is also a more widely used transfer learning model. It is developed based on the emphasis of increasing the depth of the neural network. VGG further emphasized the developing trend in terms of increased depth of networks. VGG16 has recorded the best-performing version. The innovation in this model is decreasing the filter size but increasing the depth is the necessity of building the model. Therefore, it is better to know that reducing the size of filter can only detect the small portion of image *i.e.*, better understanding of features. As the depth of the model increased then there is a need to regularize the model. Whenever the size of the dataset is small enough and the error rate is high then there is a need to add a regularization parameter to reduce the error. VGG16 model always uses the filter size of 3×3 and the dimension of max-pooling of 2×2 . The convolution operation is performed with the stride and padding of one. It is worked on a sliding window, every time sliding will give a new feature map which will be considered as input for the neural network upon which our model will be trained.

C. Experimental setup

The experiment was performed on Google Colab for training the model on the given dataset. Google Colab is the open-source IDE for running the python stuff. It automatically downloads all the libraries from the internet, externally it is not required to install any library. Nowadays, It is very popular for performing machine learning, deep learning, and computer vision experiments. These experiments require high computation and to train the model of deep learning will take many days. Deep learning gives better results if the size of the dataset will large. So, it requires GPU(Graphic Processing Unit) power for training the model in less time. Google Colab provides the GPU and TPU facility on their server. Anyone having an account on Google can use the Colab for research purposes. Keras is a high-level APIs neural network library and used for building and training the deep learning models. It is more user-friendly because its code is written in python whereas Tensorflow is an open-source library used for both high-level and low-level APIs.

Operating System: Windows10

Graphic card: Tesla K80 with 12 GB of GDDR5 VRAM, Intel Xeon Processor with two core @ 2.20 GHz

RAM: 13 GB RAM

Fine tuning: VGG16, ResNet50

1) *Splitting Data:* The dataset is divided into two parts one is the training and validation set. The training and validation ratio considered for this study is taken 90:10 means that 90 percent data is present in the training set and 10 percent in the validation set as per the size of the dataset. The description of the dataset is considered for this study is shown in Table III.

TABLE III
DIFFERENT DATASETS OF IMAGES CONSIDERED TO TRAIN THE MODEL

Sample size	Training dataset	Validation dataset
300	270	30
1500	1350	150
3000	2700	300
5000	4,050	450
6000	5,400	600

2) *Data Pre-processing:* However, we know that deep learning models require more sizes of dataset or sufficient data to train. Therefore, the technique image augmentation can be utilized to increase the dataset. After the image augmentation process, the image dataset is further pre-process using size normalization, wherein all the images are resized in a standard-sized image. In the present study, all the images are resized to 224×224. Thereafter, the images are divided/ processed into batches. We use the *batch_size = 32*, and this parameter is actually used to control the accuracy of model.

3) *Compiling model:* Compiling the model involve the choosing the optimizer, loss, and performance metrics. The hyper-parameter used to compile the model are *loss = binary_crossentropy*, *optimizer = "Adam"*, *metrics= "accuracy"*. The parameter *INIT_LR = 1 e-4*, *decay = 1 e-9* are defined in Adam optimizer which is used the learning rate of model that means that initially learning of the model starts from 0.01 and our model learned the features with the learning rate decay of 0.0001 value.

4) *Training the model:* The proposed model is trained using the initialized weights in ResNet50 and VGG16 deep neural architectures. The neural network learns from the extracted features and repeats the process of backpropagation for adjusting the weights for reducing the error. This process of adjusting the weights called fine-tuning the model. Following are the hyper-parameters used for training the model. The training of the model involves training examples, validation examples, *epochs = 50*, *steps_per_epochs=size of training data/batch_size*, and also *validation_steps=size of validation set/batch_size*.

D. Results and discussion

The metrics used for calculating the performance of the model is accuracy and loss. These measures are used both for training as well as for validation data. There are two tables which are given as ResNet50 and another is VGG16 as shown in Table IV and Table V respectively. In Table IV,

The sample size chosen for the current study are 100, 500, 1000, 1500, 2000. Firstly, our model is trained on the sample size of 300 where we achieved the accuracy of 58.51%. As we increased the sample size from 300 to 1500 there is a significant change in the accuracy of the model whereas for the sample size of 3000 the model starts to learn in a good way with the accuracy of 72.41%. After that for the sample size of 5000 only minor changes are shown in the accuracy and no change in the loss. However, for the sample size 6000, the model gives the best accuracy of 75.76%. It signifies that as we increase the sample size, the training accuracy is increased but the validation accuracy decrease after 500 sample size. Therefore, it is observed that by increasing the sample size the performance of the model on training as well as a validation set increased.

TABLE IV
RESULTS OF RESNET50 MODEL ON VARIOUS SIZE DATASET

Dataset size	Training accuracy(in %)	Validation accuracy(in %)	Percentage difference in accuracy
300	58.51	60.00	3.13
1500	63.67	80.00	22.73
3000	72.41	71.67	1.02
5000	72.42	71.78	0.88
6000	75.76	69.83	8.14

In Table V, The current study shows the experiments on the sample size of 100, 500, 1000, 1500, 2000. Firstly, our model is trained on the sample size of 300 where we achieved the accuracy of 98.15%. As we increased the sample size from 300 to 1500 there is a significant change in the accuracy of the model for the sample size 1500 the model has been given a drastic change in the accuracy with the value of 97.26% means that for the 1500 sample size accuracy decreased. By increasing the sample size from 1500 to 3000 our model gives good results than the 1500 sample size. Further, increasing dataset size on 5000 and 6000 our model better accuracy i.e., 98.67% and 99.07% respectively. Hence, It signifies that as we increase the sample size the training accuracy increases, and loss is decreased. Fig. 6 represent the comparison of loss for values during training the model. Initially, for the sample image size 300, the loss value for the ResNet50 and VGG16 is 0.54 and 0.12 respectively. As the sample image size increased from 300 to 1500 the value decreased for both the models with the value of 0.45 and 0.09 respectively. Increasing the sample image size from 1500 to 3000 again decreasing the loss value decreased with the difference value of 0.06 for the first model and 0.03 for the second model. Now the loss value for the sample image size 5000 is 0.39 and 0.04 respectively. It is observed that there is no significant change in the loss value in ResNet50 for the sample size 5000 but minor changes are observed on the VGG16 model. Again we experimented by increasing the sample image size 5000 to 6000, the loss value again decreased for both the model with the difference value of 0.04 and 0.01 for ResNet50 and VGG16 model respectively. Therefore, we can say that increasing sample size

will also decrease the loss *i.e.*, our model accurately predicting the correct class label with less value of error. The VGG16 model gives a better result compared to the ResNet50 model as shown in the Fig. 6 that there is the smooth curve for VGG16 model *i.e.*, there is a significant change in the loss value while increasing the sample image size whereas in ResNet50 model for the sample size 3000 and 5000 there is no significant change observed in the loss value.

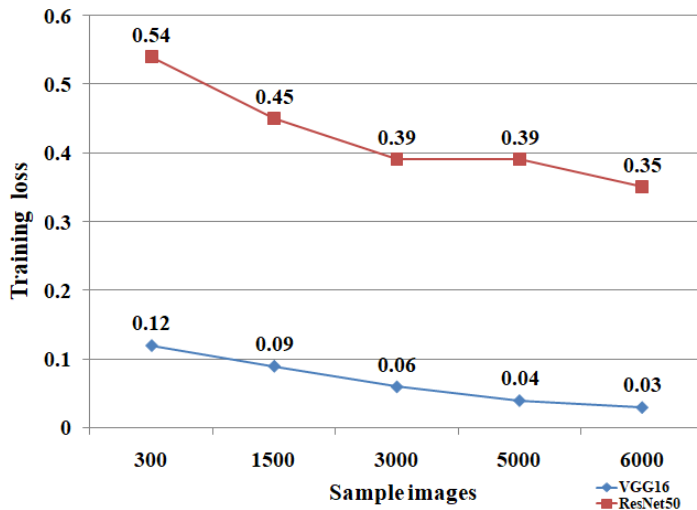


Fig. 6. Comparison of training loss on VGG16 and ResNet50 model with various sample size images

TABLE V
RESULTS OF VGG16 MODEL ON VARIOUS SIZE DATASET

Dataset size	Training accuracy(in %)	Validation accuracy(in %)	Percentage difference in accuracy
300	98.15	93.33	5.03
1500	98.26	98.00	0.75
3000	98.30	98.67	0.37
5000	98.67	98.85	0.05
6000	99.07	98.33	0.74

IV. CONCLUSION AND FUTURE WORK

In this article, the implementation and analysis of deep learning models have been attempted for the detection of wheat rust disease using two transfer learning techniques. However, rust is a common disease that occurred on wheat crops that can destroy the crop. Therefore, to cure plants from the diseases it is required to early detect the disease to get a good production. We have used the open-source dataset for this study, which is a publicly available Kaggle web source. We, further augment this dataset by using the image data augmentation technique to obtain better classification accuracy using two different deep neural network approaches (*i.e.*, ResNet50, VGG16 architecture based on Convolutional Neural Networks (CNNs)). In the experimentation, we observed that the VGG16 model performs better in comparison to the ResNet50 model. An

accuracy rate of 99.07% has been achieved with *batch_size* = 32, *optimizer* = "Adam" and *epochs* = 50. Moreover, this model will help farmers to detect diseases and prevent their crops from diseases. Further, this work can be extended by applying various feature detection techniques to remove the noise from the data and also the detected feature that will be generalized to the model.

REFERENCES

- [1] S. Kaur, S. Jindal, M. Kaur, and P. Chhuneja, "Utilization of wild species for wheat improvement using genomic approaches," in *Biotechnologies of Crop Improvement, Volume 3*, pp. 105–150, Springer, 2018.
- [2] S. Nagarajan and L. Joshi, "historical account of wheat rust epidemics in india, and their significance," *Cereal Rusts Bulletin*, 1975.
- [3] R. P. Singh, D. P. Hodson, J. Huerta-Espino, Y. Jin, P. Njau, R. Wanyera, S. A. Herrera-Foessel, and R. W. Ward, "Will stem rust destroy the world's wheat crop?," *Advances in agronomy*, vol. 98, pp. 271–309, 2008.
- [4] A. P. Roelfs, *Rust diseases of wheat: concepts and methods of disease management*. Cimmyt, 1992.
- [5] S. Nagarajan, "Intellectual property rights as an option to promote excellence in agriculture. 38th lal bahadur shastri memorial lecture," *IARI, 7th February*, 2008.
- [6] X. Chen, "Epidemiology and control of stripe rust [puccinia striiformis f. sp. tritici] on wheat," *Canadian Journal of Plant Pathology*, vol. 27, no. 3, pp. 314–337, 2005.
- [7] E. A. Milus, K. Kristensen, and M. S. Hovmøller, "Evidence for increased aggressiveness in a recent widespread strain of puccinia striiformis f. sp. tritici causing stripe rust of wheat," *Phytopathology*, vol. 99, no. 1, pp. 89–94, 2009.
- [8] T. Rumpf, A.-K. Mahlein, U. Steiner, E.-C. Oerke, H.-W. Dehne, and L. Plümer, "Early detection and classification of plant diseases with support vector machines based on hyperspectral reflectance," *Computers and electronics in agriculture*, vol. 74, no. 1, pp. 91–99, 2010.
- [9] H. Al-Hiary, S. Bani-Ahmad, M. Reyalat, M. Braik, and Z. Alrahmaneh, "Fast and accurate detection and classification of plant diseases," *International Journal of Computer Applications*, vol. 17, no. 1, pp. 31–38, 2011.
- [10] A. Akhtar, A. Khanum, S. A. Khan, and A. Shaikat, "Automated plant disease analysis (apda): performance comparison of machine learning techniques," in *2013 11th International Conference on Frontiers of Information Technology*, pp. 60–65, IEEE, 2013.
- [11] U. Mokhtar, N. El Bendary, A. E. Hassenian, E. Emary, M. A. Mahmoud, H. Hefny, and M. F. Tolba, "Svm-based detection of tomato leaves diseases," in *Intelligent Systems' 2014*, pp. 641–652, Springer, 2015.
- [12] A. Johannes, A. Picon, A. Alvarez-Gila, J. Echazarra, S. Rodriguez-Vaamonde, A. D. Navajas, and A. Ortiz-Barredo, "Automatic plant disease diagnosis using mobile capture devices, applied on a wheat use case," *Computers and electronics in agriculture*, vol. 138, pp. 200–209, 2017.
- [13] L. Xiaolong, M. Zhanhong, F. Bienvenido, Q. Feng, W. Haiguang, and J. A. Álvarez-Bermejo, "Development of automatic counting system for urediospores of wheat stripe rust based on image processing," *International Journal of Agricultural and Biological Engineering*, vol. 10, no. 5, pp. 134–143, 2017.
- [14] J. Lu, J. Hu, G. Zhao, F. Mei, and C. Zhang, "An in-field automatic wheat disease diagnosis system," *Computers and electronics in agriculture*, vol. 142, pp. 369–379, 2017.
- [15] A. Hussain, M. Ahmad, and I. A. Mughal, "Automatic disease detection in wheat crop using convolution neural network," pp. 1–4, 2018.
- [16] A. Picon, A. Alvarez-Gila, M. Seitz, A. Ortiz-Barredo, J. Echazarra, and A. Johannes, "Deep convolutional neural networks for mobile capture device-based crop disease classification in the wild," *Computers and Electronics in Agriculture*, vol. 161, pp. 280–290, 2019.
- [17] T. Vijayakumar and M. R. Vinothkanna, "Mellowness detection of dragon fruit using deep learning strategy," *Journal of Innovative Image Processing (JIIP)*, vol. 2, no. 01, pp. 35–43, 2020.
- [18] S. Shakyia, "Analysis of artificial intelligence based image classification techniques," *Journal of Innovative Image Processing (JIIP)*, vol. 2, no. 01, pp. 44–54, 2020.